COMPUTATIONAL COUPLING — Research Question Specification

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Status: planning document, superseded. This was written at the start of the project (September 2026), before any experiment was run, and it records the research plan and novelty analysis as they stood then. It is **not** a statement of findings. Several of its premises did not survive: the human H01 microscale tier was withdrawn when its connectivity graph turned out to be empty, the CoupledEventNet / neuromorphic-efficiency direction was withdrawn along with the fabricated silicon figures supporting it, and the central motif hypothesis it sets out was tested and came out with the opposite sign.

For what the project actually found, see `../info.md` and the manuscript. For what was withdrawn and why, see `../AUDIT.md`. This file is kept because the retraction is only checkable against the plan it replaced.

Project: COMPUTATIONAL COUPLING
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1. Primary Research Question (Extended Cross-Scale Formulation)

Can neural computational principles identified across biological systems remain meaningful across multiple scales of organization—from complete invertebrate nervous systems (*C. elegans*) and insect whole brains (*Drosophila*) to mammalian cortical circuits (*Mus musculus*) and human cortical micro-connectivity (H01) and whole-brain macro-connectivity (HCP)? Specifically: which structural motifs constitute functionally necessary "computational invariants", how far can these circuits be structurally remodeled under biological constraints while preserving dynamic computation, and can their abstractions produce event-driven computing architectures that achieve superior energy-delay efficiency?

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2. Biological Scale Ladder

We explicitly distinguish the different scales and modalities of biological neural organization:

1. **System A — *C. elegans*:** ~302 neurons, complete whole-animal nervous-system connectome (Cook et al. 2019, Witvliet et al. 2021).
2. **System B — *Drosophila melanogaster*:** ~139k neurons, whole adult brain connectome (FlyWire Consortium, Nature 2024).
3. **System C — *Mus musculus*:** Large cortical microcircuit (~1,000 proofread cells in Layer 2/3 of 1 mm^3 volume), combining structural connectivity + in vivo calcium activity + visual stimuli (MICrONS Consortium, Nature 2025).
4. **System D (Micro) — *Homo sapiens* (H01):** Small cortical volume (~1 mm^3 temporal cortex), resolving single neurons (~16k) and synapses (~150M), establishing human synaptic micro-connectivity and laminar E/I distributions (Shapson-Coe et al., Science 2024).
5. **System D (Macro) — *Homo sapiens* (HCP):** Whole-brain macro-scale connectivity, resolving regional tractography and functional co-activation across 180–1015 parcels (Van Essen et al. 2012, Glasser et al. 2016).

Critical Scientific Qualification: A complete human neuron-level connectome is **NOT AVAILABLE**. Whole-brain HCP data represents region-to-region tractography (voxel scale ~1.25–2 mm), while H01 represents a local 1 mm^3 micro-fragment. We do NOT treat these as directly equivalent, but as distinct hierarchical tiers.

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3. Definition: Cross-Scale Neural Computational Motif

A structural motif candidate must satisfy progressively rigorous, falsifiable criteria to qualify as a **Cross-Scale Neural Computational Motif**:

- **Level 1 (Structural Existence):** The motif exists structurally within the biological connectome graph.
- **Level 2 (Statistical Overrepresentation):** The motif occurs at a significantly higher frequency than in degree-preserving randomized null configuration models ($Z > 2.0$, $p < 0.01$).
- **Level 3 (Computational Contribution):** Embedding the motif within an event-driven dynamical system measurably improves performance on a dynamic signal-processing or learning task (e.g., coincidence detection, duration tuning, SNR).
- **Level 4 (Causal Necessity / Invariance):** Perturbing or lesioning the motif's constituent edges degrades dynamic task performance by $> 50\%$, demonstrating that it is an invariant computational bottleneck rather than redundant scaffolding.
- **Level 5 (Architectural Transferability):** A computational abstraction of the motif remains functional and advantageous when deployed outside the original biological context in an event-driven artificial neural network.
- **Level 6 (Cross-Scale Generalization):** Structurally analogous or functionally homologous configurations are detected across multiple phylogenetic levels (e.g., conserved from *C. elegans* command circuits and fly optic lobes to mouse V1 and human H01/HCP).

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4. Decomposition into Core Sub-Questions

Sub-Question 1 (Causal Motif Functionality Across Scales):

*Do conserved motifs (Coherent/Incoherent Feed-Forward Loops, Reciprocal Inhibition, Disinhibitory Triads) execute distinct, quantifiable temporal operations across invertebrate (*C. elegans*, fly) and mammalian/human cortical microcircuits (mouse V1, human H01), and do they survive Level 1–6 validation?*

Sub-Question 2 (The Structural Remodeling & Degeneracy Frontier):

Given an empirical biological neural circuit G_{orig} at each scale, how structurally divergent can a rewired circuit G' become ($\max \Delta_{struct}(G_{orig}, G')$) under strict biological constraints (Dale's principle, neurotransmitter conservation, degree caps, delay bounds) while maintaining task performance $F_{task}(G') \geq (1-\epsilon)F_{task}(G_{orig})$?

Sub-Question 3 (Event-Driven Architecture Transfer & Neuromorphic Advantage):

Can the abstracted cross-scale computational invariants be synthesized into an event-driven computing architecture (CoupledEventNet / CEN) that demonstrates Pareto superiority (accuracy vs. SynOps / latency) over conventional dense SNNs, connectome reservoirs (`conn2res`), and Neural Circuit Policies (NCPs) on temporal event-stream benchmarks?

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5. Scope Boundaries & Forbidden Claims

1. **Forbidden Over-Claim:** "The complete human brain connectome is available." (Only regional macro-connectivity [HCP] and a 1 mm^3 micro-fragment [H01] exist).
2. **Forbidden Over-Claim:** "Human brains work like fly brains" or "All brains use the exact same algorithm." (We test specific structural-dynamical invariants, acknowledging distinct ecological adaptations).
3. **Forbidden Over-Claim:** "Brain-inspired computers will replace binary digital computers." (All claims must be limited to measured metrics: SynOps, event sparsity, latency, and noise robustness on event-stream tasks).